Standard Form — Writing and Converting

AQA · KS4 Year 9–10

Name: ____________________ Class: __________ Date: __________

Write ordinary numbers in standard form ($A \times 10^n$ where $1 \le A < 10$). Convert from standard form to ordinary numbers. Understand when n is positive (large numbers) and when n is negative (small numbers).

VISUAL EXAMPLE

$$5\,600\,000 = 5.6 \times 10^{6}$$

move the point 6 places left

Standard form: $A \times 10^n$ with $1 \le A < 10$.

COMMON MISTAKES

- ✗ A must be between 1 and 10 ($1 \le A < 10$). Writing 36×10^4 is NOT standard form — it should be 3.6×10^5.
- ✗ Confusing the sign of the power — positive power = large number, negative power = small number.
- ✗ Counting the wrong number of places when moving the decimal point.
- ✗ Writing 0.003 as 3×10^3 instead of 3×10^{-3}.

METHOD

1. Identify A — move the decimal so there is one non-zero digit before it ($1 \le A < 10$).
2. Count how many places the decimal moved — this is the power n.
3. If the original number is ≥ 1, n is POSITIVE.
4. If the original number is < 1, n is NEGATIVE.
5. Write as $A \times 10^n$. Check: multiply back to verify.

WORKED EXAMPLE

Write 0.000 47 in standard form.
Step 1 — Move decimal to get 4.7 (one non-zero digit before the point).
Step 2 — The decimal moved 4 places to the right.
Step 3 — Original is less than 1, so n is negative.
Step 4 — $0.000\,47 = 4.7 \times 10^{-4}$
Check: $4.7 \times 10^{-4} = 4.7 \div 10000 = 0.000\,47$ ✓

1 Write in standard form: (2 marks)

a) 5,000 b) 340,000 c) 72,000,000 d) 810

HOW TO START

→ Put one non-zero digit before the decimal point (so $1 \le A < 10$).

→ Count how many places the point moves — that is the power.

→ A big number gives a positive power, e.g. $5\,000 = 5 \times 10^3$.

ANSWERS

(a) = ______

(b) = ______

(c) = ______

(d) = ______

METHOD 1 Move decimal so $1 \le A < 10$ | 2 Count places moved = power n | 3 Big number → positive n | 4 Small number → negative n | 5 Check by converting back

2 Write in standard form: (2 marks)

a) 0.003 b) 0.00056 c) 0.0000012 d) 0.47

HOW TO START

→ Move the point right until one non-zero digit is in front.

→ A small number (less than 1) gives a negative power, e.g. $0.003 = 3 \times 10^{-3}$.

ANSWERS

(a) = ______

(b) = ______

(c) = ______

(d) = ______

METHOD 1 Move decimal so $1 \le A < 10$ | 2 Count places moved = power n | 3 Big number → positive n | 4 Small number → negative n | 5 Check by converting back

3 Convert to ordinary numbers: (2 marks)

a) 3.2×10^{4} b) 7.05×10^{6} c) 1.8×10^{-3} d) 9.1×10^{-5}

ANSWERS

(a) = ______

(b) = ______

(c) = ______

(d) = ______

METHOD 1 Move decimal so $1 \leq A < 10$ | 2 Count places moved = power n | 3 Big number → positive n | 4 Small number → negative n | 5 Check by converting back

4 Which of these are NOT in correct standard form? Fix the ones that are wrong. (3 marks)

a) 5.6×10^{3} b) 36×10^{4} c) 0.8×10^{-2} d) 7×10^{9}

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

(d) = ____________________

METHOD 1 Move decimal so $1 \leq A < 10$ | 2 Count places moved = power n | 3 Big number → positive n | 4 Small number → negative n | 5 Check by converting back

5 The mass of a hydrogen atom is about 0.000 000 000 000 000 000 000 001 67 kg. (3 marks)

Write this in standard form.

ANSWER

Answer = ______

METHOD 1 Move decimal so $1 \leq A < 10$ | 2 Count places moved = power n | 3 Big number → positive n | 4 Small number → negative n | 5 Check by converting back

6 The distance from the Earth to the Sun is approximately 1.5×10^8 km. (3 marks)

a) Write this as an ordinary number.

b) The distance to Mars is about 2.3×10^8 km. Which planet is closer?

c) How much further is Mars than the Sun? Give your answer in standard form.

ANSWERS

(a) = ______

(b) = ______

(c) = ______

METHOD 1 Move decimal so $1 \le A < 10$ | 2 Count places moved = power n | 3 Big number → positive n | 4 Small number → negative n | 5 Check by converting back

7 Put in order, smallest first: (4 marks)

3.2×10^{-3} 8.1×10^{-4} 5.5×10^{-3} 2.9×10^{-2}

Explain your strategy.

ANSWER

Answer = ______

METHOD 1 Move decimal so $1 \leq A < 10$ | 2 Count places moved = power n | 3 Big number → positive n | 4 Small number → negative n | 5 Check by converting back

8 A bacterium is 2×10^{-6} m long. A virus is 3×10^{-8} m long. (5 marks)

a) How many times longer is the bacterium than the virus? (2 marks)
b) 500 bacteria placed end to end. Total length in standard form? (1 mark)
c) How many viruses side by side fit in 1 mm? Give in standard form. (2 marks)

ANSWERS

(a) = __________

(b) = __________

(c) = __________

METHOD 1 Move decimal so $1 \leq A < 10$ | 2 Count places moved = power n | 3 Big number → positive n | 4 Small number → negative n | 5 Check by converting back

Self-reflection

How confident do you feel with this topic now? (tick one)

☐ Not yet ☐ Getting there ☐ Confident ☐ Really confident

Which question did you find the hardest, and why?

Do I always get the sign of the power correct?

One thing I will practise before the next lesson:

Teacher key

Q1: a) 5×10^3 b) 3.4×10^5 c) 7.2×10^7 d) 8.1×10^2

Q2: a) 3×10^{-3} b) 5.6×10^{-4} c) 1.2×10^{-6} d) 4.7×10^{-1}

Q3: a) 32,000 b) 7,050,000 c) 0.0018 d) 0.000091

Q4: b) Not standard form → 3.6×10^5; c) Not standard form → 8×10^{-3}; a) and d) are correct.

Q5: 1.67×10^{-24} kg (decimal moves 24 places)

Q6: a) 150,000,000 km b) The Sun (Earth) is closer ($1.5 \times 10^8 < 2.3 \times 10^8$) c) $2.3 \times 10^8 - 1.5 \times 10^8 = 0.8 \times 10^8 = 8 \times 10^7$ km

Q7: Compare powers first: $10^{-4} < 10^{-3} < 10^{-2}$. Same power → compare A. Order: 8.1×10^{-4}, 3.2×10^{-3}, 5.5×10^{-3}, 2.9×10^{-2}

Q8: a) $(2 \times 10^{-6}) \div (3 \times 10^{-8}) = (2/3) \times 10^2 \approx 66.7$ times. b) $500 \times 2 \times 10^{-6} = 1 \times 10^{-3}$ m. c) 1 mm $= 1 \times 10^{-3}$ m; $10^{-3} \div (3 \times 10^{-8}) = (1/3) \times 10^5 \approx 3.33 \times 10^4$.